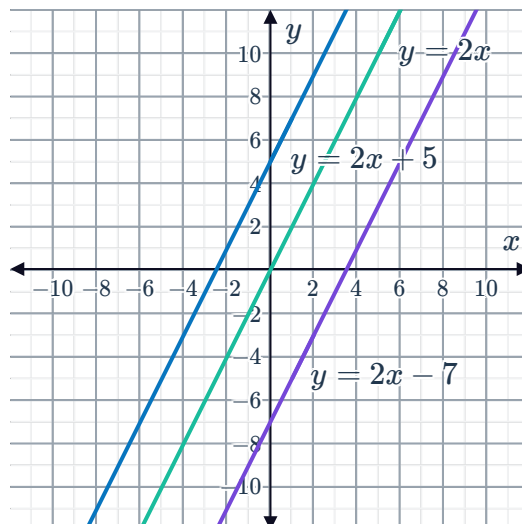


Parallel lines



- 1 The equations $y = 2x$, $y = 2x + 5$ and $y = 2x - 7$ have been graphed on the same number plane:

- a What do all of the equations have in common?
- b What do all the lines have in common?



- 2 State whether the following pairs of lines are parallel:

- a $y = -3x - 2$ and $y = -3x + 9$
- b $y = -2x - 5$ and $y = -2x - 8$
- c $y = 7x + 8$ and $y = -5x + 8$
- d $y = 4x - 1$ and $y = 4x - 6$
- e $x = 4$ and $y = 5$
- f $y = 7x - 5$ and $y = -7x + 6$
- g $y = 3x - 5$ and $y = -\frac{1}{3}x + 6$
- h $y = -8x - 2$ and $9x + 7$

- 3 State whether the following lines are parallel to $y = 7x + 3$:

- a $y = 7x - 3$
- b $y = 6x + 3$
- c $y = 7x + 4$
- d $y = 7x$
- e $y = 6x + 4$
- f $y = -7x + 3$
- g $y = 3x + 7$
- h $y = -3x - 7$

- 4 State whether the following lines are parallel to $y = -3x + 2$:

- a $y = 3x$
- b $-3y - x = 5$
- c $y = -10 - 3x$
- d $y + 3x = 7$

- 5 State whether the following lines are parallel to $y = 9x + 2$:

- a $y = 9x$
- b $y = -9x + 5$
- c $y = -9x + 2$
- d $y = 9x - 2$

- 6 Consider the lines $L_1 : 5x + 8y - 10 = 0$ and $L_2 : y = -\frac{5x}{8} + 8$.

- a Find the gradient of line L_1 .
- b Find the gradient of line L_2 .
- c Are the two lines parallel?

- 7 Consider the line $y = 2x + 2$. If every point on the line is shifted 2 units up, find the equation of the new line.
- 8 Every point on a particular line is shifted 6 units down. The resulting line has equation $y = 2x - 2$. Find the equation of the original line.
- 9 Find the equation of a line described by the following information:
- Parallel to the x -axis and passes through $(-10, 2)$.
 - Parallel to the y -axis and passes through $(-7, 2)$.
 - Parallel to the line $y = -3x - 8$ and cuts the y -axis at -4 .
 - Parallel to the line $y = 8x - 3$ and cuts the y -axis at 5 .
 - Parallel to the line $y = -2x + 9$ and passes through the point $(-3, 1)$.
- 10 If the following pairs of lines are parallel, find the value of b :
- $$\begin{aligned} 0 &= 5x - 9y + 6 \\ 0 &= bx + 3y - 2 \end{aligned}$$
 - $$\begin{aligned} y &= bx + 4 \\ 2y &= \frac{4}{5}x - 7 \end{aligned}$$
- 11 The line L_1 goes through $(3, 2)$ and $(-2, 4)$.
- Find the gradient of line L_1 .
 - Find the equation of the line that has a y -intercept of 1 and is parallel to line L_1 .
- 12 The line L_1 goes through $(-9, 10)$ and $(2, -1)$.
- Find the gradient of line L_1 .
 - Find the equation of the line that passes through $(10, 8)$ and is parallel to line L_1 .
- 13 The line L_1 passes through the point $(9, -5)$ and is parallel to the line $y = -5x + 2$.
- Find the gradient of line L_1 .
 - Find the equation of line L_1 .

Perpendicular lines

- 14 Describe what it means for two lines to be perpendicular.
- 15 If two lines are perpendicular, state the product of their gradients.

16 State whether the following pairs of lines are perpendicular:



a $y = x + 1$ and $y = x - 1$

b $y = 3x - 7$ and $y = -3x + 6$

c $y = -3x + 6$ and $y = \frac{x}{3} + 7$

d $y = \frac{2x}{3} + 4$ and $y = -1 - \frac{3x}{2}$

e $y = -\frac{x}{3} - 9$ and $y = -(3x - 27)$

f $y = 3x - 5$ and $y = -\frac{1}{3}x + 6$

g $y = -\frac{2x}{3} + 1$ and $y = -1 + \frac{3x}{2}$

h $y = \frac{3x}{5} + 6$ and $y = \frac{-5x - 30}{-3}$

17 Given that the following pairs of lines are perpendicular, calculate the value of m :

a $5x + 4y + 8 = 0$
 $mx + 9y - 8 = 0$

b $-6x + 6y - 8 = 0$
 $mx + 10y + 8 = 0$

18 Find the gradient of the line perpendicular to the following lines:

a Line with gradient 6

b $y = 6x + 3$

c $y = \frac{x}{8} + 3$

d $y = 8 - x$

19 Consider the lines $L_1: 8x = -7y + 3$ and $L_2: 7y - 8x - 7 = 0$.

a Find the gradient of line L_1 .

b Find the gradient of line L_2 .

c Are the two lines perpendicular?

20 Find the equation of a line described by the following information:

a Perpendicular to the x -axis and passes through $(-8, -1)$.

b Perpendicular to the y -axis and passes through $(-8, -8)$.

c Perpendicular to $y = -\frac{x}{2} + 5$, and goes through the point $(0, 6)$.

d Perpendicular to $y = 6x + 10$, and has the same y -intercept.

e Perpendicular to the line that passes through $(1, 1)$ and $(3, 13)$, and has a y -intercept of $(0, -4)$.

21 The line L_1 is perpendicular to $y = 5x - 4$ and cuts the y -axis at 1.

a Find the gradient of line L_1 .

b Find the equation of line L_1 .

22 The line L_1 is perpendicular to $y = 10x +$  passes through the point of intersection of the lines $y = x + 5$ and $8x - 10y + 60 = 0$.

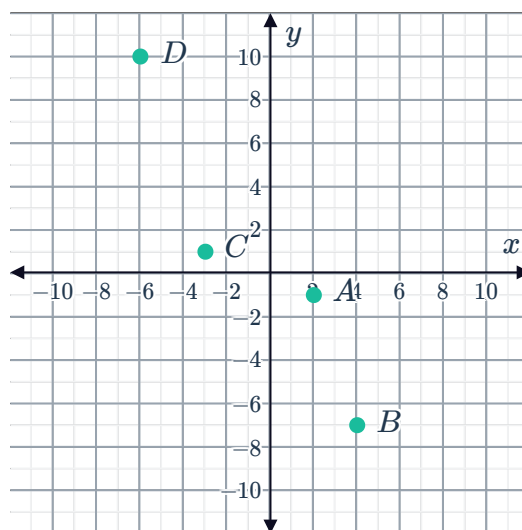
- Find the gradient of line L_1 .
- Find the point of intersection of $y = x + 5$ and $8x - 10y + 60 = 0$.
- Find the equation of the perpendicular line L_1 .

Applications

23 Consider the following points on the number plane:

- Point $A(2, -1)$
- Point $B(4, -7)$
- Point $C(-3, 1)$
- Point $D(-6, 10)$

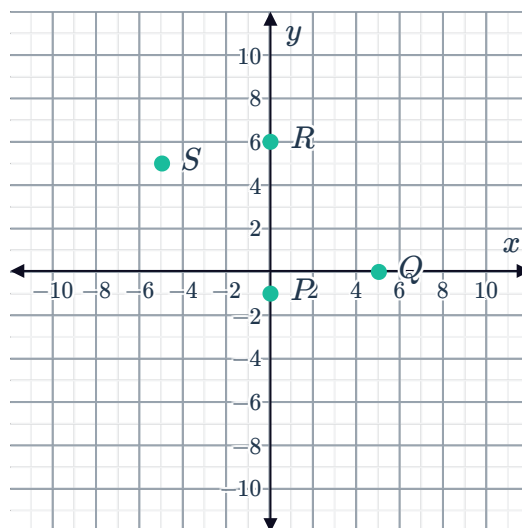
- Find the gradient of the line AB .
- Find the gradient of the line CD .
- Is the line CD parallel to AB ?



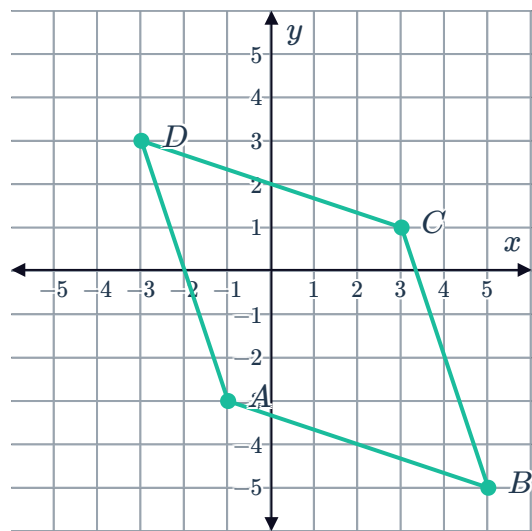
24 Consider the following points on the number plane:

- Point $P(0, -1)$
- Point $Q(5, 0)$
- Point $R(0, 6)$
- Point $S(-5, 5)$

- Find the gradient of PQ .
- Find the gradient of RS .
- Are PQ and RS parallel?
- Are QR and PS parallel?
- What type of quadrilateral is $PQRS$?



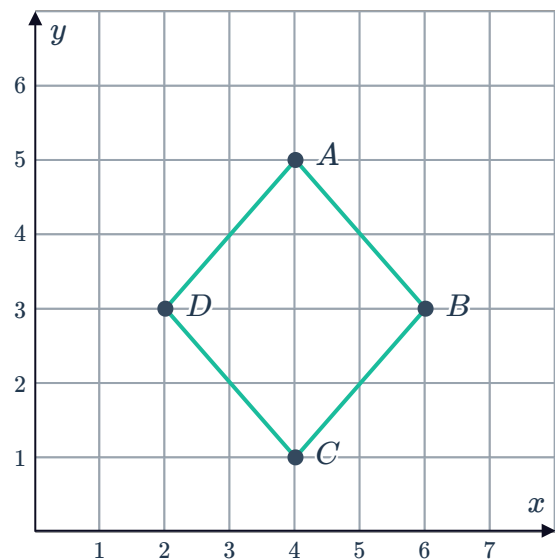
25 Consider the rhombus $ABCD$ on the number plane:



- a Find the gradients of its diagonals.
 - i Gradient of AC
 - ii Gradient of BD
- b Are the diagonals of the rhombus perpendicular? Explain your answer.

26 The four vertices of quadrilateral $ABCD$ have been plotted on the number plane:

- a Find the gradient of the following sides:
 - i AB
 - ii BC
 - iii DC
 - iv AD
- b Given that side $AB = BC$, determine the type of quadrilateral described by the four points.



27 Determine whether the following sets of points are collinear:

- a Point $A(-4, 3)$, Point $B(-2, 7)$ and Point $C(-7, -3)$.
- b Point $A(-3, 1)$, Point $B(-2, 6)$ and Point $C(0, 14)$.
- c Point $A(-2, 4)$, Point $B(2, 2)$ and Point $C(4, 1)$.